

EACKNATH SIR'S
40 PRACTICE QUESTIONS
ON STRING

Question 1 — Reverse a String

Problem Statement

Write a program to reverse a given string.

Input

A string S.

Output

Print the reversed string.

Examples

Example 1:

Input: hello

Output: olleh

Example 2:

Input: I am iron man

Output: nam nori ma I

Question 2 — Check if a String is Palindrome

Problem Statement

Given a string, check whether it is a palindrome. A string is a palindrome if it reads the same forward and backward.

Input

A string S.

Output

Print Palindrome if S is a palindrome; otherwise print Not Palindrome.

Examples

Example 1:

Input: ABCDCBA

Output: Palindrome

Example 2:

Input: TAKE U FORWARD

Output: Not Palindrome

Question 3 — Count Vowels, Consonants and Spaces

Problem Statement

Given a string, count the number of vowels, consonants, and spaces in it.

Input

A string S.

Output

Print the count of vowels, consonants, and spaces.

Examples

Example 1:

Input: Take u forward is Awesome

Output: Vowels: 10, Consonants: 11, White spaces: 4

Example 2:

Input: India won the cricket match

Output: Vowels: 8, Consonants: 15, White spaces: 4

Question 4 — Find the ASCII Value of a Character

Problem Statement

Given a character, find its ASCII value.

Input

A single character C.

Output

Print the ASCII value of C.

Examples

Example 1:

Input: A

Output: 65

Example 2:

Input: e

Output: 101

Question 5 — Remove All Vowels from a String

Problem Statement

Given a string, remove all vowels from the string.

Input

A string S.

Output

Print the string after removing vowels.

Examples

Example 1:

Input: take u forward

Output: tk frwr

Example 2:

Input: I am very happy today

Output: m vry hppy tdy

Question 6 — Remove Spaces from a String

Problem Statement

Given a string, remove all spaces from the string.

Input

A string S.

Output

Print the string after removing spaces.

Examples

Example 1:

Input: take U forward

Output: takeUforward

Example 2:

Input: How are you doing

Output: Howareyoudoing

Question 7 — Remove Characters Except Alphabets

Problem Statement

Given a string, remove every character except alphabetic characters.

Input

A string S containing letters, digits, spaces, and/or special characters.

Output

Print only the alphabetic characters, preserving their original order.

Examples

Example 1:

Input: take12% *&u; ^\$#forward

Output: takeufoward

Example 2:

Input: Java&C;++

Output: JavaC

Question 8 — Remove Brackets from an Algebraic Expression

Problem Statement

Given an algebraic expression, remove all bracket characters without changing the order of the remaining characters.

Input

An algebraic expression S.

Output

Print the expression after removing brackets.

Examples

Example 1:

Input: a+((b-c)+d)

Output: a+b-c+d

Example 2:

Input: (((a-b))+c)

Output: a-b+c

Question 9 — Sum of the Numbers in a String

Problem Statement

Given a string, calculate the sum of all numbers in it. Multiple consecutive digits are considered one number.

Input

A string S containing letters and digits.

Output

Print the sum of all numbers present in S.

Examples

Example 1:

Input: 123xyz

Output: 123

Example 2:

Input: 1xyz23

Output: 24

Question 10 — Capitalize First and Last Character of Each Word

Problem Statement

Given a string, capitalize the first and last character of every word.

Input

A string S containing words separated by spaces.

Output

Print the modified string.

Examples

Example 1:

Input: take u forward is awesome

Output: TakE U ForwarD IS AwesomE

Example 2:

Input: I am a boy

Output: I AM A BoY

Question 11 — Calculate Frequency of Characters

Problem Statement

Given a string, calculate the frequency of every character in the string.

Input

A string S.

Output

Print each character and its frequency.

Examples

Example 1:

Input: takeuforward

Output: a2 d1 e1 f1 k1 o1 r2 t1 u1 w1

Example 2:

Input: articles

Output: a1 c1 e1 i1 l1 r1 s1 t1

Question 12 — Find Non-Repeating Characters

Problem Statement

Given a string, print the characters that occur exactly once.

Input

A string S.

Output

Print all non-repeating characters in their original order.

Examples

Example 1:

Input: google

Output: l e

Example 2:

Input: yahoo

Output: y a h

Question 13 — Check if Two Strings are Anagrams

Problem Statement

Given two strings, check whether they are anagrams of each other.

Input

Two strings S1 and S2.

Output

Print true if they are anagrams; otherwise print false.

Examples

Example 1:

Input: CAT, ACT

Output: true

Example 2:

Input: RULES, LESRT

Output: false

Question 14 — Find the Maximum Occurring Character

Problem Statement

Given a string, return a character that occurs the maximum number of times. If multiple characters tie, return any one of them.

Input

A string S.

Output

Print one character having maximum frequency.

Examples

Example 1:

Input: takeuforward

Output: a

Example 2:

Input: apple

Output: p

Question 15 — Remove All Duplicate Characters

Problem Statement

Given a string, remove repeated occurrences so that each character appears only once, preserving the first occurrence order.

Input

A string S.

Output

Print the resulting string.

Examples

Example 1:

Input: bcabc

Output: bca

Example 2:

Input: cbacdcbc

Output: cbad

Question 16 — Print All Duplicate Characters with Counts

Problem Statement

Given a string containing lowercase letters a to z, print characters that occur more than once along with their counts.

Input

A lowercase string S.

Output

Print every character whose frequency is greater than 1 and its count. If none exists, print an empty result.

Examples

Example 1:

Input: sinstriiintng

Output: i:4, n:3, s:2, t:2

Example 2:

Input: abcdefg

Output: []

Question 17 — Remove Characters from First String Present in Second

Problem Statement

Given two strings, remove from the first string every character that is present in the second string.

Input

Two strings S1 and S2.

Output

Print the remaining characters of S1.

Examples

Example 1:

Input: abcdef, cefz

Output: abd

Example 2:

Input: xyzpw, lmno

Output: xyzpw

Question 18 — Change Every Letter to the Next Alphabet

Problem Statement

Replace every alphabetic character with the next letter in the alphabet. After z, continue with a; preserve letter case.

Input

A string S.

Output

Print the transformed string.

Examples

Example 1:

Input: abcdxyz

Output: bcdeyza

Example 2:

Input: Java

Output: Kbw b

Question 19 — Find the Largest Word in a String

Problem Statement

Given a string, find the longest word. If multiple words have the same maximum length, return the first one.

Input

A string containing words separated by spaces.

Output

Print the longest word.

Examples

Example 1:

Input: Google Doc

Output: Google

Example 2:

Input: Microsoft Teams

Output: Microsoft

Question 20 — Sort Characters in a String

Problem Statement

Sort the alphabetic characters of a string in ascending order.

Input

A string containing alphabetic characters.

Output

Print the sorted string.

Examples

Example 1:

Input: zxcbg

Output: bcgxz

Example 2:

Input: edcba

Output: abcde

Question 21 — Count the Number of Words

Problem Statement

Given a string, count the number of words. A word is a sequence of non-space characters.

Input

A string S.

Output

Print the number of words.

Examples

Example 1:

Input: Hello

Output: 1

Example 2:

Input: Hi there

Output: 2

Question 22 — Find the Word with Highest Repeated-Letter Frequency

Problem Statement

Find the word containing a letter with the highest frequency among all words. If no word contains a repeated letter, print -1.

Input

A string containing words separated by spaces.

Output

Print the first word with the highest repeated-letter frequency, or -1.

Examples

Example 1:

Input: abcdefghij google microsoft

Output: google

Example 2:

Input: cameron blue

Output: -1

Question 23 — Change Case of Each Character

Problem Statement

Change lowercase letters to uppercase and uppercase letters to lowercase. Leave non-alphabetic characters unchanged.

Input

A string S.

Output

Print the string after changing the case of every alphabetic character.

Examples

Example 1:

Input: java

Output: JAVa

Example 2:

Input: take u forward IS Awesome

Output: TAKE U FORWARD is aWESOME

Question 24 — Concatenate Two Strings

Problem Statement

Concatenate two strings and form one combined string while preserving their original order.

Input

Two strings S1 and S2.

Output

Print the concatenated string.

Examples

Example 1:

Input: hello, world

Output: helloworld

Question 25 — Find the Position of a Substring

Problem Statement

Given a text and a pattern, find the first occurrence of the pattern in the text.

Input

Two strings: text and pattern.

Output

Print the starting index of the first occurrence, or -1 if the pattern is not present.

Examples

Example 1:

Input: takeuforward, forward

Output: 5

Example 2:

Input: hello, az

Output: -1

Question 26 — Reverse Words in a String

Problem Statement

Reverse the order of words in a string. The output must contain exactly one space between words and no leading or trailing spaces.

Input

A string S containing letters, digits, and spaces.

Output

Print the words in reverse order with single spaces.

Examples

Example 1:

Input: welcome to the jungle

Output: jungle the to welcome

Example 2:

Input: amazing coding skills

Output: skills coding amazing

Question 27 — Count Digits, Letters and Special Characters

Problem Statement

Given a string, count how many characters are digits, alphabetic characters, and special characters.

Input

A string S.

Output

Print the three counts.

Examples

Example 1:

Input: Hello123!

Output: Letters: 5, Digits: 3, Special: 1

Example 2:

Input: A@1#

Output: Letters: 1, Digits: 1, Special: 2

Question 28 — Count Uppercase and Lowercase Characters

Problem Statement

Given a string, count the number of uppercase and lowercase alphabetic characters.

Input

A string S.

Output

Print the uppercase and lowercase counts.

Examples

Example 1:

Input: Hello World

Output: Uppercase: 2, Lowercase: 8

Example 2:

Input: ABCdef

Output: Uppercase: 3, Lowercase: 3

Question 29 — Remove Duplicate Words from a String

Problem Statement

Given a sentence, remove repeated words while preserving the first occurrence of each word.

Input

A sentence S.

Output

Print the sentence containing each word only once.

Examples

Example 1:

Input: this is is a test test

Output: this is a test

Example 2:

Input: hello hello world

Output: hello world

Question 30 — Count Occurrences of a Given Character

Problem Statement

Given a string and a character, count how many times the character occurs in the string.

Input

A string S and a character C.

Output

Print the number of occurrences of C.

Examples

Example 1:

Input: banana, a

Output: 3

Example 2:

Input: hello, l

Output: 2

Question 31 — Replace Every Space with a Hyphen

Problem Statement

Given a string, replace every space character with a hyphen.

Input

A string S.

Output

Print the modified string.

Examples

Example 1:

Input: hello world

Output: hello-world

Example 2:

Input: Python is easy

Output: Python-is-easy

Question 32 — Check if a String Contains Only Digits

Problem Statement

Given a string, determine whether every character in the string is a digit.

Input

A string S.

Output

Print True if all characters are digits; otherwise print False.

Examples

Example 1:

Input: 12345

Output: True

Example 2:

Input: 123a45

Output: False

Question 33 — Check if a String Contains Only Alphabets

Problem Statement

Given a string, determine whether every character is an alphabetic character.

Input

A string S.

Output

Print True if all characters are alphabets; otherwise print False.

Examples

Example 1:

Input: Hello

Output: True

Example 2:

Input: Hello123

Output: False

Question 34 — Find the First Non-Repeating Character

Problem Statement

Given a string, find the first character whose frequency is exactly one.

Input

A string S.

Output

Print the first non-repeating character, or -1 if none exists.

Examples

Example 1:

Input: swiss

Output: w

Example 2:

Input: aabbcc

Output: -1

Question 35 — Find the First Repeating Character

Problem Statement

Given a string, find the first character that appears more than once.

Input

A string S.

Output

Print the first repeating character, or -1 if no character repeats.

Examples

Example 1:

Input: swiss

Output: s

Example 2:

Input: abcdef

Output: -1

Question 36 — Remove a Given Character from a String

Problem Statement

Given a string and a character, remove all occurrences of that character.

Input

A string S and a character C.

Output

Print the string after removing C.

Examples

Example 1:

Input: banana, a

Output: bnn

Example 2:

Input: hello, l

Output: heo

Question 37 — Count Occurrences of a Substring

Problem Statement

Given a string and a substring, count how many times the substring occurs without overlapping occurrences.

Input

Two strings: text and pattern.

Output

Print the number of occurrences of pattern in text.

Examples

Example 1:

Input: aaaa, aa

Output: 2

Example 2:

Input: hello hello, hello

Output: 2

Question 38 — Check if One String is a Rotation of Another

Problem Statement

Given two strings, check whether the second string can be obtained by rotating the first string.

Input

Two strings S1 and S2.

Output

Print True if S2 is a rotation of S1; otherwise print False.

Examples

Example 1:

Input: ABCD, CDAB

Output: True

Example 2:

Input: ABCD, ACBD

Output: False

Question 39 — Find the Shortest Word in a String

Problem Statement

Given a string containing words, find the shortest word. If multiple words have the same minimum length, return the first one.

Input

A string containing words separated by spaces.

Output

Print the shortest word.

Examples

Example 1:

Input: I love Python

Output: I

Example 2:

Input: code in Java

Output: in

Question 40 — Check if a String is a Valid Identifier

Problem Statement

Given a string, determine whether it can be used as a valid Python identifier.

Input

A string S.

Output

Print Valid if S is a valid Python identifier; otherwise print Invalid.

Examples

Example 1:

Input: student_name

Output: Valid

Example 2:

Input: 123student

Output: Invalid